Empirical Methods for Law and Economics

Course syllabus – October 2026

Degree programme	European Doctorate in Law and Economics (EDLE)
University	University of Bologna
Language	English
Instructor	Marco Rosso
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Lecture notes	Teams or Teaching Material

Course description

This introductory doctoral course provides the statistical and econometric tools needed to read, evaluate and carry out empirical and experimental research on legal questions. No previous statistics or econometrics is assumed; some mathematical knowledge is required. Every meeting combines concepts with practical work in Stata, and every application comes from Law and Economics: contract enforcement, sanctions, prosecutorial independence, legal institutions and legal texts.

Format. Eight teaching meetings of 3 hours each (including at least one break), followed by two 3-hour meetings for presentations and assessment: 30 hours in total.

Learning outcomes

By the end of the course, students will be able to:

1. describe variables and distributions, stating units, denominators and the unit of observation;
2. interpret sampling uncertainty, standard errors, confidence intervals and basic statistical tests;
3. estimate and interpret simple and multiple OLS regressions, including indicator (dummy) variables;
4. discuss regression assumptions and distinguish association from causation;
5. explain random assignment and the basic design of experiments;
6. construct a simple quantitative measure from legal texts and assess its validity;
7. keep a reproducible analysis in a Stata do-file;
8. write an accurate and constructive referee report on an empirical paper.

Programme

Meeting	Concepts	Practice and paper discussion
1	Research questions, unit of observation, measurement, variable types, descriptive statistics, missing values	Stata: open, inspect, tabulate and summarize World Bank data on contract enforcement; histograms
2	Sampling, standard deviation vs standard error, sampling distributions, confidence intervals	Stata: supplied simulation. Paper: Gneezy and Rustichini (2000)

Meeting	Concepts	Practice and paper discussion
3	Relationships between variables, correlation, hypothesis tests for means and proportions	Stata: comparisons and tests; association vs causation
4	Simple OLS: coefficients, predictions, residuals, indicator variables	Stata: fit and interpret models. Paper: Voigt and Wulf (2019), first reading
5	Multiple OLS: controls, indicators, model building. Anatomy of a referee report	Stata: compare models; conditional associations. Good and weak referee comments on Voigt and Wulf (2019)
6	OLS assumptions, uncertainty, interpreting empirical results	Stata: model assessment, changing samples. Paper: Voigt and Wulf (2019), second reading, written as a referee comment
7	Experimental economics: random assignment, incentives, internal and external validity	Classroom vignette and simulated data; evaluating an experimental design
8	Introduction to text as data: building a measure from legal texts; validity of text-based measures	Stata: dictionary counts on a legal corpus. Paper: Gentzkow, Kelly and Taddy (2019); feedback on draft comments
9–10	Individual presentations and discussion of the agreed papers	

Required readings

Paper discussions take place in meetings 2, 4, 6 and 8; meeting 5 uses a short guide to referee reports. Read only the indicated parts (30–45 minutes each) and bring short notes on the questions in the reading schedule distributed with the materials of each week. Reading notes are not graded.

Meeting	Paper	What to read
2	Gneezy, U. and Rustichini, A. (2000). A fine is a price. *Journal of Legal Studies* 29(1): 1–17. Download the paper	Abstract; Sections I–II; Table 1 (p. 6) and Figure 1 (p. 7). Skip the formal tests and the model.
4	Voigt, S. and Wulf, A. J. (2019). What makes prosecutors independent? Analysing the institutional determinants of prosecutorial independence. *Journal of Institutional Economics* 15(1): 99–120 (first online 2017). Download the paper	Abstract; Sections 1 and 5; Section 7 and the first two paragraphs of Section 8 (pp. 112–113); Table 1; column (1) of Table 2.
5	Berk, J. B., Harvey, C. R. and Hirshleifer, D. (2017). How to write an effective referee report and improve the scientific review process. *Journal of Economic Perspectives* 31(1): 231–244. Download the paper	The whole article (short). Suggested: Thomson (2011), Chapter 4, "Writing Referee Reports" (pp. 151–166).
6	Voigt and Wulf (2019), second reading	Section 8 in full; all columns of Table 2 and footnotes.
8	Gentzkow, M., Kelly, B. and Taddy, M. (2019). Text as data. *Journal of Economic Literature* 57(3): 535–574. Download the paper	Sections 1 and 2; Section 3.5.2 (model validation and interpretation). Skip the statistical methods in Section 3.

Extra, optional (meeting 8). For students who want to apply text methods beyond this introduction: Python for Economists, Block III (text representation, sentiment analysis, machine learning for text). Not required and not assessed.

Assessment

Assessment is individual: a written referee report (70%) and an oral presentation with questions (30%).

Choice of the work. Each student chooses a working paper, or a seminar or workshop presentation, in agreement with the instructor. It must contain at least one empirical result that can be discussed with the tools of this course.

Written report (70%). 1,500–2,000 words in English, excluding references. It must contain:

1. an accurate summary: research question, contribution, data, unit of observation, sample, method and main findings;
2. the explanation of one central empirical result, citing the exact table, figure or slide, with magnitude, units, comparison, sample size and uncertainty where reported;
3. two or three specific substantive comments, each stating the issue, why it matters for a conclusion, and a feasible clarification or check;
4. an overall assessment with at least one strength and the main limitation.

For a seminar, distinguish information that was not available to you from evidence that the study lacks a feature. A replication and advanced econometric methods are not required.

Oral presentation and discussion (30%). In meetings 9–10, each student gives a presentation of about 20 minutes, followed by about 15 minutes of questions from the class. Present the question, one central result, a strength and your most important comment. Be ready to explain the statistical concepts involved and one Stata output from the course practice. The 30% is split into two components:

- *Presentation and answers (20%)*: clarity, correct use of the statistical concepts, and quality of the answers to questions.
- *Questions to colleagues (10%)*: in each presentation other than your own, you are expected to ask or contribute at least one question. The instructor records them and grades only your best three. A good question is accurate, specific to the paper's evidence (a table, a sample, an identification step) and constructive. The number of questions does not matter beyond the three that count; generic or repetitive questions earn no points.

Milestones (formative, not graded):

- meeting 3: the chosen paper or seminar is agreed with the instructor;
- meeting 5: one-page outline (question, data, method);
- meeting 7: draft comments;
- meeting 8: feedback and revision in class;
- before meeting 9: submission of the final report, by a common deadline communicated by the instructor.

Cite all sources consulted and disclose any assistance, including AI tools, according to the University's rules. Every student must be able to explain their report orally.

Practical information

- Bring a laptop with Stata installed (version 16 or later; University of Bologna licence or personal licence).
- Keep all course files in the `Stata/do/` folder. In Stata, set the working directory to that folder (`cd "path/to/Stata/do"`): it contains `data/`, `exercises/` and `00_setup.do`.

- Student do-files are worksheets completed in class; supplied demonstration scripts require no programming knowledge. Simulated data are always labelled as such.
- Slides, worksheets and reading questions are made available before each weekly block; solutions are shared after each lecture.

Reference books

No textbook is required; specific sections are indicated week by week.

- Wooldridge, J. M., *Introductory Econometrics: A Modern Approach*, Cengage (6th or 7th edition).
- Angrist, J. D. and Pischke, J.-S., *Mastering 'Metrics: The Path from Cause to Effect*, Princeton University Press (2015).
- Studenmund, A. H., *Using Econometrics: A Practical Guide*, Pearson (7th edition, 2017).
- Acock, A. C., *A Gentle Introduction to Stata*, Stata Press (any recent edition).
- Thomson, W., *A Guide for the Young Economist*, MIT Press (2nd edition, 2011).
- Optional, free: Diez, D., Çetinkaya-Rundel, M. and Barr, C. D., *OpenIntro Statistics*, 4th edition, https://www.openintro.org/book/os/. Useful background for meetings 1–3.